Thermal radiation of Leslie's cube and a black body

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We tested the T^4-dependency of thermic radiation of a body. To examine different surfaces we used Leslie's cube. A model black body was also used for this test. With this model black body we also observed the angular distribution of the emitted thermic radiation, which follows Lambert's law. We measured the infrared transmission coefficients of window glass $\tau = (1.72 \pm 0.59)\,\%$, of p-doped silicon $\tau = (44.78 \pm 0.54)\,\%$ and of a NaCl cristal $\tau = (75.32 \pm 0.57)\,\%$.

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1 Introduction

Every body that has a temperature above absolute zero emits and absorbs energy in the form of thermic radiation. Depending on the temperature of the object, this electromagnetic radiation has an intensity maximum in a certain wavelength range. If a body is slightly hotter than room temperature, its radiation maximum is in the infrared range, which is why infrared radiation is usually referred to as thermic radiation. The intensity maximum of the sun, the strongest source of thermic radiation in our area, is in the ultraviolet range. The intensity distribution of thermic radiation of an object is described by Planck's law of radiation. Max Planck was one of the first physicists to recognize, while he was dealing with the course of the spectral energy density of thermic radiation, that electromagnetic radiation is not continuous, but rather exchanged in discrete units. This discovery laid the foundation for quantum physics. Still several properties of thermic radiation can be described classically.

In the following experiment the Stefan-Boltzmann law will be examined using Leslie's cube and a nearly ideal black body. In addition, the infrared transmittance of three different substances is determined and the Lambert law will be examined.

2 Theory of thermic radiation

Every object emits thermal radiation in form of electromagnetic radiation and absorbs thermic radiation from its environment. If the emission and absorbtion is equal, the object is in thermal equilibrium with its environment. In this equilibruim Kirchhoff´s law of radiation (1) holds, where ϵ is the absorptance of the object and P_s is the emited power of radiation of a so called black body radiator.

$$P = \epsilon P_\mathrm{s} \tag{1}$$

This leads to the finding that an object that has a higher emission also has a higher absorbtion.

A black body is an ideal radiator. Following eq. (1), it has an absorptance of $\epsilon = 1$. The energy density ρ of its radiation of a frequency ν only depends on its temperature T. It follows Planck´s law of radiation:

$$\rho(\nu, T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{\mathrm{e}^{h\nu/(k_\mathrm{B}T)} - 1} \tag{2}$$

$h = 6,62607015 \cdot 10^{-34}\mathrm{Js}$ is Planck´s constant, $k_\mathrm{B} = 1,380649 \cdot 10^{-23}\,\frac{\mathrm{J}}{\mathrm{K}}$ is Boltzmanns constant and $c = 2,9970923458 \cdot 10^8\,\frac{\mathrm{m}}{\mathrm{s}}$ [1] is the speed of light in vacuum. This ideal thermal radiation has its maximum intensity at the wavelegth

$$\lambda_\mathrm{max} = 2897,8\mu\mathrm{m} \cdot \frac{1}{\mathrm{T/K}} \tag{3}$$

following Wien´s displacement law.
One can derive the previously known Stefan-Boltzmann law (4) from Planck´s law of radiation.

$$P = \sigma A T^4 \tag{4}$$

P is the total emitted power of radiation, A the radiating area and $\sigma = 5,670374419 \cdot 10^{-8}\,\frac{\mathrm{W}}{\mathrm{m^2K^4}}$ [1]

the Stefan-Boltzmann constant. One can see, that the power does not depend on the symmetry of the radiating object but only on its radiating area and its temperature.

A black body is in good approximation a Lambert-radiator, following Lambert´s cosine law (5). This law describes the angular distribution of the radiated intensity.

$$I = I_0 \cdot \cos\theta \tag{5}$$

θ is the angle between the line of sight and the surface normal of the radiating area. A non-Lambert radiator usually has a higher intensity in the direction of the surface normal.

3 Stefan-Boltzmann law

3.1 Leslie's cube

The experimental setup for the examination of the thermic radiation of Leslie's cube is shown in Fig. 1. We use Leslie's cube to study the influence of different surface properties on the thermic radiation af an object with a constant temperature. The cube has a black and a white couloured side and two nickel plated sides. One of the nickel plated surfaces is matt, the other one is polished. The cube is rotatably mounted. The temperature of the cube is regulated by a *Lauda RE104* thermostat. The thermic radiation of the cube is measured by a *Kipp & Zonen* thermopile. The distance between the radiating surface and the thermopile is $\approx 12,5\,\mathrm{cm}$. A filter can be placed between the thermopile and the cube. The thermo voltage measured by *Agilent 34405 A* is directly proportional to the emitted radiation power due to the Seebeck effect.

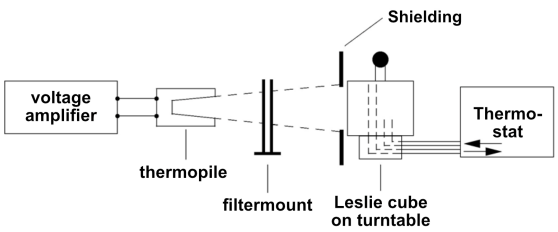

Fig. 1: Experimental setup to examine thermic radiation with Leslie's cube. [2]

We measure the thermic radiation of all four sides of Leslie's cube without a filter at temperatures between 15 °C and 85 °C.

In Fig. 2 the thermo voltage of all four sides is plotted against the temperature. One can see that the black and the white side of Leslie's cube radiate

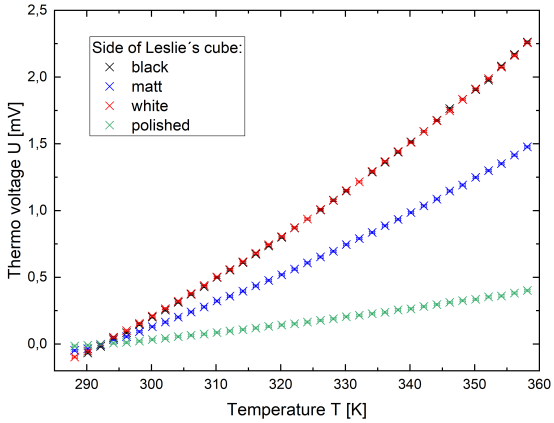

Fig. 2: The measured thermo voltage of all sides of Leslie's cube proportional to their thermic radiation. With a linearisation for low temperatures we get the refence temperature T_0.

very similarly. This indicates that both sides have similar emission behaviour in the infrared range. This shows that the emission and absorbtion behavior of objects differ in different ranges of the electromagnetic spectrum as they behave differently in the visible range.

The nickel plated sides of Leslie's cube have a lower emission than the black and the white coloured sides. The polished side has a high refelctivity and with that a low emission rate. The matt side has a lower reflectivity than the polished one due to its rough surface. With that comes a higher emission rate for the matt side.

To test the T^4 dependency of the Stefan-Boltzmann law we fitted the function

$$U = a\,(T^b - T_0^b) \tag{6}$$

to all four sides. Via a linearisation of the six data points with the lowest tempereature for all sides the reference temperature gets determined to $T_0 = (292.2 \pm 5.7)\,\mathrm{K}$. This is the temperature where there is no thermo voltage in the thermopile. This coincides with the temperature of the environment $T_{\mathrm{env}} = (293.2 \pm 1.0)\,\mathrm{K}$. With this reference tempereature T_0 we get the exponents b of the eq. (6) for all four sides of Leslie's cube, shown in Tab. 1.

None of the determined exponents coincedes with the theoretically expected value of $b = 4$ following the Stefan-Boltzmann law (4). Especially for the black coloured and the white coloured surface one would expect the experiment to follow the Stefan-Boltzmann law. Due to Kirhoff's law, one expects the exponent for both nickel plated surfaces lower than the ideal value $b = 4$. One would also not expect that the exponent of the matt nickel plated surface is the closest to the expected value.

One attempt to explain the deveations from the

surface	exponent b
polished nickel plated	3.835 ± 0.089
matt nickel plated	3.929 ± 0.041
white	3.643 ± 0.018
black	3.797 ± 0.045

Tab. 1: Determined exponent b for all four different sides of Leslie's cube.

theory could be that the surface of Leslie's cube facing towards the thermopile was not exatly perpendicular to the thermopile. This would cause a drop in the intensity following Lambert's law (see section 5). If the surface is a non-Lambert radiator this drop would be even more significant than if it follows Lambert's law. This could cause the too low exponent b. An other reason is the inaccuracy of the reference temperature T_0 and the proportionality factor a in the fit function (6). Also one has to mention that the Stefan-Boltzmann law only describes the behavior of an ideal black body radiator. This is just an idealisation of real radiating objects and can not exist in reality.

3.2 Black body

To get as near to an ideal black body radiator as possible, we use an electric oven that heats a cavity with quasi ideally absorbing walls. The thermic radiation gets emitted through a small hole and two water cooled aperture. With a thermopile (analogue to section 3.1) we measure the thermo voltage as shown in Fig. 3. The setup infront of the model black body can rotated in respect to the black body.

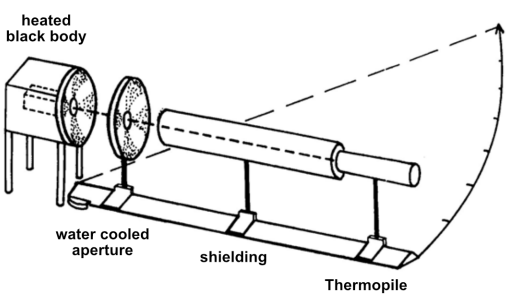

Fig. 3: Experimental setup to examine thermic radiation with a black body. [2]

The measurement and the evaluation is analogue to section 3.1, we only replace Leslie's cube with the model black body. The orientation for this measurement is 0 °. While the oven heats the cavity continuously, we measure the thermo voltage with a *Agilent 34405 A* in steps of 5 °C. This data is shown in Fig. 4.

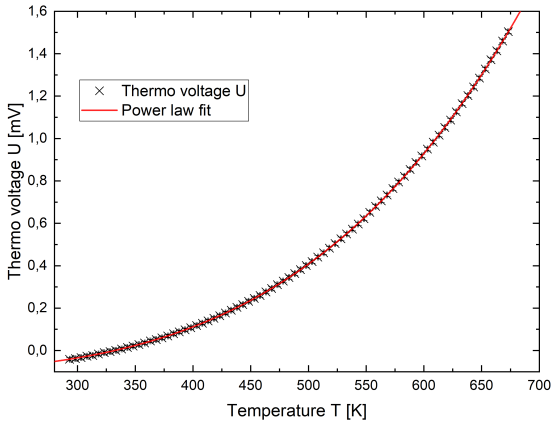

Fig. 4: Thermo voltage proportional to the thermic radiation of a model black body in dependence of its temerature.

With a fit analogue to section 3.1, we get an exponent $b_{\text{black body}} = 3.8324 \pm 0.0041$. This value does not coincede with the theoretically expected value $b = 4$. The heating of the environment through the measurement with Leslie's cube and through the strong radiation of the sun during the forenoon could be a reason for this deviation from the Stefan-Boltzmann law.

4 Infrared transmittance

The experimental setup according to Fig. 1 is used. In order to measure the infrared transmittance of substances, three different filters are inserted into the holder. Leslie´s cube is kept at a constant temperature $T = (348.15 \pm 0.40)$ K while the black side is used. Wien´s displacement law (3) is used to check whether the maximum of the emitted thermic radiation is in the infrared range. The calculated maximum leads to the value:

$$\lambda_{\max} = (8.3234 \pm 0.0096)\,\mu\text{m} \tag{7}$$

This value lies in the infrared range. For the measured room temperature $T_0 = (293.2 \pm 1.0)$ K the following maximum can be determined:

$$\lambda_{\max} = (9.883 \pm 0.034)\,\mu\text{m} \tag{8}$$

In order to get a reference value for the radiation, which is emitted from the black side of the cube, it is measured without a filter at first. To determine the zero point voltage U_0 of the thermoelectric pile a nickel plated disk, which blocks all of the emitted thermic radiation of the black side, is inserted into the holder. A 3 mm thick piece of window glass, a 0.5 mm

thick p-doped silicon plate and a 5 mm thick NaCl crystal are inserted successivley into the holder and the thermic radiation after penetrating the filter is measured. The transmission coefficient τ is examined using (9).

$$\tau = \frac{U - U_0}{U_{\text{black}} - U_0} \tag{9}$$

The resulting transmission coefficients τ of the three different substances are listed in Tab. 2.

filter material	transmission coefficient τ [%]
window glass	1.72 ± 0.59
silicon	44.78 ± 0.54
NaCl	75.32 ± 0.57

Tab. 2: Transmission coefficients of differnt substances.

Due to its structural design, window glass has the lowest transmissivity. Window glass is the amorphous form of SiO_2. It therefore only has a local order and an irregular pattern. Due to this disordered structure, SiO_2 has several vibrational and rotational degrees of freedom. Therefore a lot of infrared radiation is absorbed and only little radiation can penetrate the glass without interaction. The silicon filter and the NaCl filter both show a higher transmissivity. Silicon has a diamond structure and NaCl has a fcc structure. Therefore these two materials have a long-range order, which leads to less vibrational and rotational degrees of freedom. Therefore the lattice have less opportunities to take up vibrations and absorb energy when hit by radiation. One reason for the higher transmissivity of NaCl could be the difference in the phonon band gap of the two materials. Due to the ionic bonds in the NaCl crystal, this phonon band gap is larger than that of silicon with a covalent bond. Therefore there is a larger range of photon energy that cannot be absorbed by the lattice, resulting in a larger transmission coefficient for NaCl.

5 Lambert's law

In order to determine the relation between the detected radiation and its angle of inclination θ, the experimental setup shown in Fig. 3 is used. The thermopile is mounted rotatable about the ideal black body radiator and the deflection angle θ can be measured. To measure this relation and to check whether it follows Lambert's law, the temperature of the black body must remain constant. This was difficult due to the inertia of the regulating transformer, which supplies the thermic radiator. The temperature cannot be set precisely, but can only be increased and decreased alternately and measured at a specific temperature during a pass. The temperature $T = (587 \pm 3)\,\text{K}$ is adjusted in which the uncertainty occurs from this inertia. in order to compare the measured course with the theoretical course of Lambert's law, U_θ normed by $U_{\theta=0}$ is plotted against $\cos\theta$ (Fig. 5).

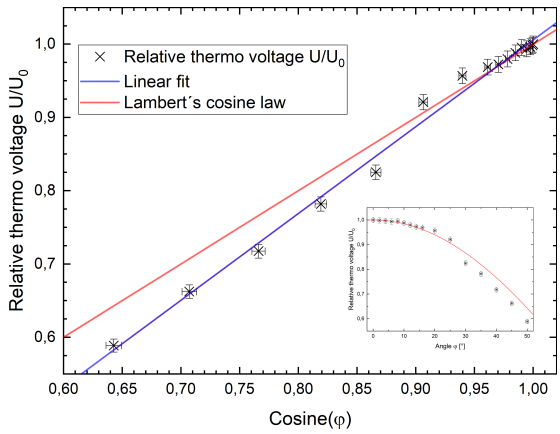

Fig. 5: The normed thermo voltage is plotted against $\cos\theta$ and a linear fit is applied to the data. A straight line with slope $b = 1$ is plotted for comparison with the theoretical course. In addition, U_θ is plotted against the angle θ, which shows the expected cosine curve.

Compared to the theoretical curve with the slope of $b_\text{t} = 1$, taking into account the not exactly constant temperature, the measured line with $b_\text{m} = 1.184 \pm 0.026$ agrees well. The larger deviation of the measured thermo voltage at small $\cos\theta$ values results from a larger deviation of the set temperature in these measurements.

6 Conclusion

We examined the power law relation between the emitted thermic radiation and the temperature of a body. For this we used Leslie's cube and a model black body. For the black coloured side of Leslie's cube we measured a exponent $b = 3.797 \pm 0.045$, the white coloured side leads to an exponent $b = 3.643 \pm 0.018$. The analysis of the emitted thermic radiation of the two nickel plated surfaces leads to $b = 3.929 \pm 0.041$ for the matt side and $b = 3.835 \pm 0.089$ for the polished surface. For the thermic radiation emitted by a model black body we measured $b = 3.8324 \pm 0.0041$. These determined values do not coincide with the theoretically expected exponent $b = 4$.
We examined the infrared transmittance of three materials. We calculated the maximum of the emitted radiation to $\lambda_\text{max} = (8.3234 \pm 0.0096)\,\mu\text{m}$ to ensure that it is in the infrared range. The transmission coefficient of window glass was determined to

$\tau = (1.72 \pm 0.59)\,\%$, p-doped silicon has a transmission of $\tau = (44.78 \pm 0.54)\,\%$ and a NaCl cristal transmits $\tau = (75.32 \pm 0.57)\,\%$ of the radiation.
We also covered the relation between the detected radiation and the angle of inclination. The intensity of the radiation follows a cosine as discribed by Lambert's law.

References

[1] P. J. Mohr, D. B. Newell, and B. N. Taylor: "CODATA recommended values of the fundamental physical constants: 2014", Rev. Mod. Phys. 88, 035009 (2016), last visited 9th March, 2023 `https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=920686`

[2] Physikalisches Grungpraktikum, Universität Würzburg, Modul C2, Versuch 33, "Währmestrahlung„